

Computer Security 4321- Milestone 2 Literature Paper
Prompt Injection Defenses in Large Language Models (LLMs)
Large Language Models for Computer Security
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ABSTRACT

Security is an important aspect when it comes to artificial intelligence or just technological systems in general. Including security in technological systems is a key factor as it keeps everything within the system monitored and secure. This literature review introduces our main tool and two comparable tools that we are going to be basing our research on. First, our primary tool is the Prompt Injection tool. It is a scanner for a system that reviews user input to ensure it is safe before letting it be fully processed by the system. In addition to the Prompt Injection tool there will be two comparable tools that will also be included in our research: the Rebuff tool, and Vigil tool. The Rebuff and Vigil tools are also classified as review type scanners but the Rebuff tool provides a risk score of how malicious an input is, and the Vigil tool reviews inputs using YARA rules to identify malicious code patterns and report them back to the user.

ARTICLE 1

Firstly, we examined an article titled “Prompt Injection Detection in LLM Integrated Applications”. This article discusses how large language models are being integrated in new applications and how they not only provide upsides, but also potential risks to systems. The article then focuses on prompt injection detection, where a further explanation of the tool, certain aspects of the tool, and benefits that come with LLMs integrated with said tool. They then provide an example of an LLM with prompt injection detection implemented into it. The system runs a security check through a series of combined tests that include going through a list of terms that are banned, embedding techniques, and a BERT model. While comparing this tool to our own we can see that they implemented a significantly greater number of tests than we implemented on our tool to fully ensure every possible scenario is caught. Other than the number of tests, it seems that our tools have the same layout process when it comes to testing and decision making before being processed to the LLM.

ARTICLE 2

The second article that was found is titled “Enhancing Security in LLM Applications: A Performance Evaluation of Early Detection Systems”. This text goes over the topic of how certain prompt injection defense tools are unreliable against certain strategies used within prompt injection attacks. Within the article it also shows how the writers have studied, tested, and reviewed a number of tools that were used to prevent prompt injection attacks where the test results provide how tools reacted to these attacks. When compared to our tool, the tests were conducted on the same three tools we tested in part 1 of this milestone, the article reports testing of the PromptInjection, Rebuff, and Vigil tool and draws conclusions based on the findings. With our tests we showed that the 3 tools performed adequately against basic prompt injection attacks, but as stated in the article different tests were tried which led to the conclusion that Vigil and Rebuff could not detect data leakages and that they are better for basic tests.

ARTICLE 3

Our third article was titled “Defending Language Models Against Image-Based Prompt Attacks via User-Provided Specifications”. It examines security for MLLM systems, like chatbots, but highlights the limited protection when it comes to image injections, which represents a potential attack vector. Overall this paper focuses on implementing a defense tool for image-based prompt injection attacks for multi-modal large language models. Their tool combines input validation and a prompt injection detection to ensure submitted images are safe. The approach is similar to our PromptInjection tool, but instead of being used to review inputted text it is stated that they use the tool to review inputted images that may have malicious text encoded into the image, which is the main difference between their tool and ours.

ARTICLE 4

The title of article number four was “Evaluating the efficacy of LLM Safety Solutions: The Palit Benchmark Dataset” and the article discusses the evaluation of various security tools. By generating malicious prompts and testing them against various prompt injection defense tools

they were able to classify which tools performed the best and which ones failed to fully identify the malicious content. When compared to our project, this approach is very similar to our project's testing phase as we also took fake prompts and tested them against the PromptInjection, Vigil, and Rebuff tool, although the key difference is that the writers of the article tested more tools than just the three we used.

ARTICLE 5

The fifth article is titled “SPML: A DSL for Defending Language Models Against Prompt Attacks”. This article discusses a “domain specific language” that monitors user inputs and refines prompts before they are processed by an LLM chatbot. This tool was made to protect the chatbot from malicious inputs ensuring that prompts contain no malicious intent; otherwise, they are rejected. When comparing, the tool reviewed in this article performs similarly to our PromptInjection tool where both tools review user inputs, processes input before evaluation, and rejects input text if it is identified as malicious.

CONCLUSION

In conclusion, the literature review allowed us to analyze and compare multiple security tools and approaches related to Prompt Injection. The articles examined included tools and methods similar to those used in our project, such as Prompt Injection, Rebuff, and Vigil. Through this comparison, we were able to understand the strengths, limitations, and different implementations of these tools. Overall, this literature review highlighted the importance of implementing reliable protections against Prompt Injection attacks.

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keywords: {Computer

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